

1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page. **Hint:** If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

Steps to Complete the Practical

1. Connect your laptop/PC to your **home Wi-Fi**.
2. Open **Command Prompt** (Win + R → type cmd → Enter).
3. Type:

ipconfig

4. Find **Default Gateway** under your Wi-Fi adapter. It may look like 192.168.1.1.
5. Open Chrome/Edge and enter that address in the address bar, for example:

http://192.168.1.1

6. The **router login page** should appear.
7. **Do not share your router password.** Take a screenshot showing the login page (you can hide the username if it contains personal information).
8. Add the screenshot to your assignment with the caption: **“Home Wi-Fi Router Admin Login Page.”**

If 192.168.1.1 or 192.168.0.1 doesn't work, use the exact Default Gateway shown by ipconfig.

2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

If you are using a **Windows laptop**, follow these steps:

1. Connect your laptop to your **Wi-Fi**.
2. Open **Command Prompt**.
3. Type:

ipconfig

4. Under **Wireless LAN adapter Wi-Fi**, note these values:

Network Detail	Example
IP Address	192.168.17.133
Subnet Mask	255.255.255.0
Default Gateway	192.168.17.1

Assignment Answer

Device: Laptop

Connection: Wi-Fi

- **IP Address:** 192.168.17.133 (*example — use your actual value*)
- **Subnet Mask:** 255.255.255.0 (*example*)
- **Default Gateway:** 192.168.17.1 (*example*)

3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.
Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

Practical: Set a Static IP and Test Internet

Steps on Windows 11

1. Connect your laptop to your **Wi-Fi**.
2. Open **Settings → Network & internet → Wi-Fi**.
3. Select your connected Wi-Fi network.
4. Find **IP assignment → Edit**.
5. Change **Automatic (DHCP)** to **Manual**.
6. Turn on **IPv4**.
7. Enter values based on your router. For example, if your router is 192.168.1.1:

Setting	Example
IP Address	192.168.1.50
Subnet Prefix Length	24
Gateway	192.168.1.1
Preferred DNS	8.8.8.8
Alternate DNS	1.1.1.1

8. Click **Save**.
9. Open a browser and visit a website to test the Internet connection.
10. You can also open CMD and run:

ping 8.8.8.8

If you receive replies, the connection is working.

Revert to DHCP

After testing, go back to **IP assignment** → **Edit** → **Automatic (DHCP)** → **Save**.

Assignment Result

Static IP: 192.168.1.50

Gateway: 192.168.1.1

Internet Access: Working / Not Working (*write your actual result*)

Final Setting: Reverted to **Automatic (DHCP)**.

4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

1. Log in to your router's admin panel.
2. Look for **LAN Settings**, **DHCP Server**, or **Network Settings**.
3. Find **DHCP Start IP** and **DHCP End IP**.
4. Note the range shown by your router.

Example:

- **DHCP Start IP:** 192.168.1.100
- **DHCP End IP:** 192.168.1.200
- **Current DHCP Range:** 192.168.1.100 – 192.168.1.200

What happens if two devices have the same IP?

If two devices are assigned the **same IP address**, an **IP address conflict** occurs. The devices may lose network/Internet connectivity or experience unstable communication because the network cannot reliably determine which device should receive the traffic.